Do Financial LLM Agents Discover New Alpha? A Fidelity Audit, Disclosed-Formula Study, and Prompt-Decision Replay

Anonymous Author(s)

Abstract

Financial LLM papers often report profitable factors or portfolios, but cross-paper attribution is meaningful only when the tested object preserves the source mechanism. We screen 98 works and strictly audit an existing 50-strategy common-task study. Its grades are A0/B0/C15/D33/U2; 46 strategies combine ranked Jensen–Kelly–Pedersen (JKP) characteristics and 47 share essentially the same monthly long–short construction. We therefore demote that ladder to a construction diagnostic. The primary formula sample instead exhaustively includes the three evaluator-valid seeds in a pinned QuantEvolver release. All three preserve the released expressions, executable DSL semantics, pairwise missing-data rule, forward-close return, and equal-mean top/bottom-quintile evaluator; only cadence, universe, and horizon change. A fail-closed audit reports 3/3 strict grade B components (100%). Across four matched benchmarks, their median annualized out-of-window residual ranges from −0.4009% to +1.2250%; one row is Holm-positive under FF3 and none under the broad JKP model. Separately, we replay all 190 disclosed GuruAgents prompt cells through a current 2026 OpenRouter-served `openai/gpt-4o` alias and form 24 costed paths. For the 12 paths sharing 33 factor months, adding JKP betting-against-beta to official FF5 plus momentum reduces median annualized alpha from 6.12% to 2.87% and attenuates 11 paths, while one Buffett path remains Holm-positive. The 100% result applies only to disclosed components, not native agents or training pipelines. Broad claims about agent alpha still require native outputs, adequate histories, costs, and multiplicity control.

CCS Concepts

• **Computing methodologies → Multi-agent systems**; • **Applied computing → Economics**; • **General and reference →** *Empirical studies.*

Keywords

financial agents, alpha, asset pricing, factor zoo, reproducibility, multiple testing

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1 Introduction

LLM-agent papers increasingly report profitable factors, superior trading performance, or investment alpha [1, 4, 7, 9, 10]. Positive returns can coexist with exposures to low-risk, momentum, quality, or valuation factors, so broad spanning is useful. It is not useful, however, if the tested "agent strategy" is a researcher-created combination of the same characteristics used as controls.

That distinction changes this paper's evidentiary order. A strict audit of the prior 50-strategy layer assigns no A or B grades: A=0, B=0, C=15, D=33, and U=2. Forty-six strategies combine ranked JKP characteristics, 47 use essentially the same monthly long–short portfolio, at least 39 had a closer public formula, prompt, algorithm, or execution rule, and the mappings were not outcome-blind or independently second-coded. The resulting ladder answers a narrow construction question—whether JKP factors span researcher-authored JKP-like composites—but not whether they span the papers' native agents. We retain it only as a transparent final-section diagnostic.

The main study asks two smaller questions with stronger source anchors. For GuruAgents [3], does replaying the disclosed prompts and deterministic tool observations through the current GPT-4o alias produce economically distinct portfolios, and how do those paths load on standard factors? For QuantEvolver [10], what happens when we exhaustively execute every evaluator-valid released seed with the pinned DSL and portfolio evaluator on an adapted monthly U.S. panel? A separate 12-formula exercise covering current EFS [5], QuantEvolver, Alpha-Jungle [6], and QuantAgent [8] remains a mixed-fidelity diagnostic outside the primary denominator. Neither primary layer reproduces a complete native system, model training process, search trajectory, or published performance table.

Contributions. First, we convert a damaging proxy-validity concern into an explicit audit with fixed A/B/C/D/U meanings and denominators. Second, we provide a prompt-decision component replay with cell-level outputs, holdings, turnover, costs, and matched factor attribution. Third, we define a source-exhaustive three-component census whose released expressions, evaluator semantics, missing-data rule, forward returns, and portfolio rule all pass a fail-closed grade-B gate. Fourth, we state the exact multiplicity family for every Holm adjustment and quarantine post-hoc or mixed-fidelity proxy evidence from the main claims.

Takeaway. The paper does not show that JKP132 absorbs financial LLM agents. It shows how attribution changes when the empirical object progresses from low-fidelity proxy composites to strictly disclosed formula and prompt-decision components, while making the 100% denominator explicit.

2 Evidence Design

2.1 Corpus and strict proxy-fidelity audit

The cutoff-bounded screen covers database inception through 2 August 2026 and contains 98 distinct works: 69 formula-discovery or trading papers are retained and 29 benchmarks, audits, tools, or mechanistic comparators are excluded with recorded reasons. The search did not preserve raw hit rankings, so it is a structured evidence map rather than a claim of exhaustive coverage. Unavailable native agents are not assigned zero returns.

The audit grades the *empirical object*, not a paper's scientific quality. Grade A requires a faithful paper/system replication. Grade B requires the same disclosed formula, prompt-decision component, or return stream; only cadence, universe, holding period, weights, and transaction costs may change. Grade C preserves a recognizable idea but materially changes its signal. Grade D is materially inconsistent with the source. Grade U means that the cited source was unavailable for formula-level review. Applied to the prior 50 common-task strategies, the fixed counts are A0/B0/C15/D33/U2. Even the previously named "source-grounded" subset grades C8/D5. No row reproduces native-agent output or supports a paper-level performance claim.

This audit is logically prior to spanning. Because 46 of the 50 strategies combine ranked JKP characteristics and 47 use essentially the same monthly long–short rule, regressing those returns on JKP factors is partly a construction check. We preserve the estimates and provenance in Section 7, but exclude them from the main evidence for agent alpha.

2.2 Three faithful disclosed components

The primary denominator is the exhaustive set of evaluator-valid example seeds in QuantEvolver commit `4eb0e78842138ada` (the table records the full hash). Three seeds are valid; the fourth is explicitly labeled as an invalid unknown-operator seed and is rejected by the source evaluator. Table 1 records the three counted expressions. Source-file hashes pin the candidate list, DSL evaluator, and cross-sectional rank-IC evaluator. This scope is fixed from the release rather than selected from realized returns.

The implementation preserves the expression trees and executable numerical semantics: absolute-price close, close-derived returns, $\log(|\text{volume}| + 10^{-8})$, population standard deviation, 10^{-8} safeguards, and the 240-bar warm-up. It also preserves the source evaluator's pairwise score/forward-return deletion before ranking, $q = \max(1, \lfloor 0.2n \rfloor)$ tail size, and equal-mean top-minus-bottom portfolio. The outcome is the next available-bar close return. There is no missing-return imputation, reranking, substitution, turnover adjustment, or transaction cost in the counted path.

Only grade-B mechanical adaptations are made: bars are monthly rather than daily, the universe is the 1,000 largest positive-market-equity U.S. common stocks, and the holding horizon is one monthly bar. Each component supplies 305 return observations (915 total), with 184,596 formation holdings. Six holdings use a nonconsecutive next available bar exactly as the source evaluator specifies. Factor alignment leaves 270 common months; the first 120 train each rolling model and the remaining 150, August 2009 through January 2022, form the matched evaluation window. The gate is therefore 3/3 strict grade B (100%) for disclosed evaluator components, not for QuantEvolver's LLM search, training, or native system.

2.3 GuruAgents prompt-decision component replay

GuruAgents is evaluated separately because its public repository names GPT-4o, supplies temperature-zero prompts and deterministic finance-tool observations, and archives portfolios. It does not pin the original provider or model snapshot. We execute 190/190 agent–quarter–mode cells through a current 2026 OpenRouter-served `openai/gpt-4o` alias at a total API cost of \$9.91; this preserves the disclosed interface but is not an exact model replication. The *archived-final* mode supplies the completed source tool transcript before the final allocation decision; *tool-routing* begins from the exact system prompt and user request and lets the model call the corresponding archived tools. Mean ticker-set Jaccard agreement with the authors' holdings is 0.569 and 0.607, respectively. The replay is therefore a disclosed prompt-decision component, not a reconstruction of source data collection or the authors' archived return path.

The replay produces 24 paths: both modes for five agents and an equal-weight ensemble in two source archives. Quarter-end formations execute at the first subsequent trading close, weights drift between quarterly rebalances, dividend-adjusted closes determine gross returns, and the paper-declared 1 bp times one-way traded notional determines net returns. Matched attribution restricts every model to the 12 long-archive paths and the same 33 realized months, April 2022–December 2024.

2.4 Attribution and multiplicity families

Formula attribution is out of window. At each evaluation month, CAPM, FF3, and FF5 plus momentum use OLS loadings estimated on the preceding 120 months. The broad FF5-plus-momentum-plus-JKP132 model leaves the six base-factor slopes unpenalized and selects a ridge penalty separately for each component using the final 24 months inside that rolling training window. Annualized mean residuals are then evaluated on the same 150 months. Newey–West statistics use maximum lag four. Holm adjustment is across the three faithful components *separately within each benchmark*; it is not across the four benchmark specifications.

For GuruAgents, the identified OLS ladder uses official CAPM, FF3, and FF5 plus momentum, then adds either JKP BAB alone or a fixed seven-factor low-risk block: BAB, conventional and Dimson beta, downside beta, CAPM idiosyncratic volatility, realized volatility, and quality-safety. Newey–West statistics use maximum lag three. Holm adjustment is across the 12 replay paths *separately within each benchmark*; unavailable paths are omitted, not imputed, and the adjustment is not across benchmark specifications. Unrestricted JKP132 OLS needs 134 parameters including the intercept and is unidentified with 33 observations. We label a pre-2022 factor-only PCA compression as outcome-blind and an all-JKP nested leave-one-month-out ridge fit as exploratory and noncausal; neither enlarges the identified OLS claim.

3 Results

3.1 Faithful disclosed components

The fail-closed gate accepts all three counted components as strict grade B. It checks source identity, expression, evaluator semantics,

Table 1: Primary counted sample: exhaustive evaluator-valid seeds in the pinned QuantEvolver release. Every row is strict grade B.

Seed	Source expression	Preserved source evaluator	Grade
Q1	div(ts_mean(returns(60)), ts_std(returns(60)))	Released DSL numerics; pairwise score/forward-return deletion; equal-mean top/bottom quintiles	B
Q2	−zscore(last(close(120)), close(120))	Released DSL numerics; pairwise score/forward-return deletion; equal-mean top/bottom quintiles	B
Q3	corr(returns(60), log(|volume(60)| + 10^{-8}))	Released DSL numerics; pairwise score/forward-return deletion; equal-mean top/bottom quintiles	B

Source: https://github.com/QuantLLM/QuantEvolver at commit `4eb0e78842138ada5334349585b114ad923564e8`. The release also contains `seed_0004`, explicitly named `bad_unknown_op`; the source evaluator rejects it, so it is not an empirical candidate. Cadence, universe, and horizon change; no native agent or training pipeline is replicated.

missing-data handling, forward return, and portfolio construction. Any conditional grade, source-hash mismatch, changed candidate census, imputation, or altered portfolio rule makes the validator fail. Holdings independently reconstruct every return in the full-evidence package.

Across the three-row family, median annualized out-of-window residual is +1.2250% under CAPM (3/3 positive; 0/3 Holm-positive), +0.2549% under FF3 (2/3; 1/3), −0.4009% under FF5 plus momentum (1/3; 0/3), and +0.6713% under FF5 plus momentum plus JKP132 (2/3; 0/3). Q2, the released price-z-score reversal, is Holm-positive only under FF3: annualized residual 6.9508%, HAC $t = 2.6009$, and Holm $p = 0.0279$. Its nominal broad-model $p = 0.0460$ becomes 0.1381 after the declared three-component correction.

The matched attribution uses the same return paths and 150 evaluation months at every benchmark step. The results are a faithful test of three disclosed evaluator components under allowed cadence, universe, and horizon adaptations. They do not reproduce QuantEvolver's search trajectory or establish native-system performance. The older 12-formula bundle remains useful as a sensitivity diagnostic, but its five B-conditional and four C-conditional rows are excluded from this 100% denominator.

3.2 Prompt replay: paired BAB attribution

The prompt replay shows that the language-model decision layer is empirically material. The two replay modes reproduce only 10 and 9 of 95 archived ticker orderings exactly, and their mean ticker-set agreement is well below one. Across replay–author common samples, however, mean return correlation is 0.952 because both use the same archived signals and compiled Nasdaq source universe. The replay is a stronger test of the disclosed decision component, not an independent out-of-sample validation.

Figure 1 pairs each of the 12 matched replay paths before and after adding BAB. Official FF5 plus momentum leaves median alpha of 6.12%. Adding BAB lowers it to 2.87%: 11 of 12 alphas fall and 11 of 12 BAB loadings are positive. The larger low-risk block leaves median alpha of 3.50%. This is attenuation, not universal absorption.

One archived-final Buffett path remains Holm-positive under FF5 plus momentum, BAB, the low-risk block, and the outcome-blind compressed-JKP specification. The exploratory all-JKP ridge leaves 0 Holm-positive paths. The two facts are compatible: 33 months identify the small OLS models but not an unrestricted 133-factor regression, while the ridge diagnostic is noncausal and model-dependent. We therefore neither declare a universal surviving agent alpha nor promote the ridge result to proof of full absorption.

Table 2: Prompt-replayed GuruAgents factor attribution. Every row uses the same 12 paths and 33 realized months, net of 1-bp one-way costs.

Benchmark	Median α	$\alpha > 0$	Holm +
Official CAPM	7.30%	12/12	0/12
Official FF3	6.23%	12/12	0/12
Official FF5+Mom	6.12%	12/12	1/12
+ JKP BAB	2.87%	11/12	1/12
+ JKP low-risk block	3.50%	11/12	1/12

The BAB row adds only JKP `betabab_1260d`. The low-risk block adds BAB, conventional and Dimson beta, downside beta, idiosyncratic and realized volatility, and quality-safety returns. Holm adjustment is across the 12 replay paths separately within each benchmark; there is no adjustment across benchmark specifications. Full JKP132 OLS is unidentified in 33 months.

3.3 The strict audit changes the estimand

The A0/B0/C15/D33/U2 result rules out the original paper-level interpretation before any return regression is run. A C-grade return can test whether a recognizable motif has value under our implementation; a D-grade return cannot be attributed to the cited paper. Pooling either group as "replicated agents" would make the source label do more work than the empirical object supports. We therefore do not use the 50-strategy ladder to estimate an agent-alpha prevalence rate, rank papers, or infer that agents rediscover JKP factors.

The audit also explains why apparent broad-factor absorption is unsurprising in that layer. Most scores are explicit functions of JKP columns, while almost every return path applies the same cross-sectional rank and long–short rule. Their nearest-factor correlations are properties of those constructions. Section 7 reports the full ladder so that this negative design lesson remains reproducible rather than disappearing from the record.

3.4 What the source papers establish

Among the 38 mapped papers whose full text we verified, 37 report no asset-pricing regression, 1 reports loadings without a factor-adjusted intercept, none reports factor-adjusted alpha, and none uses JKP132. Source evaluations instead emphasize predictive scores, cumulative or excess returns, Sharpe ratios, buy-and-hold comparisons, agent baselines, or factor loadings. Those are legitimate task-specific outcomes. They simply answer a different question from incremental return spanning.

4 Interpretation and Limits

The evidence hierarchy is the result. The legacy 50-strategy family is transparent and statistically reproducible, but its strict fidelity

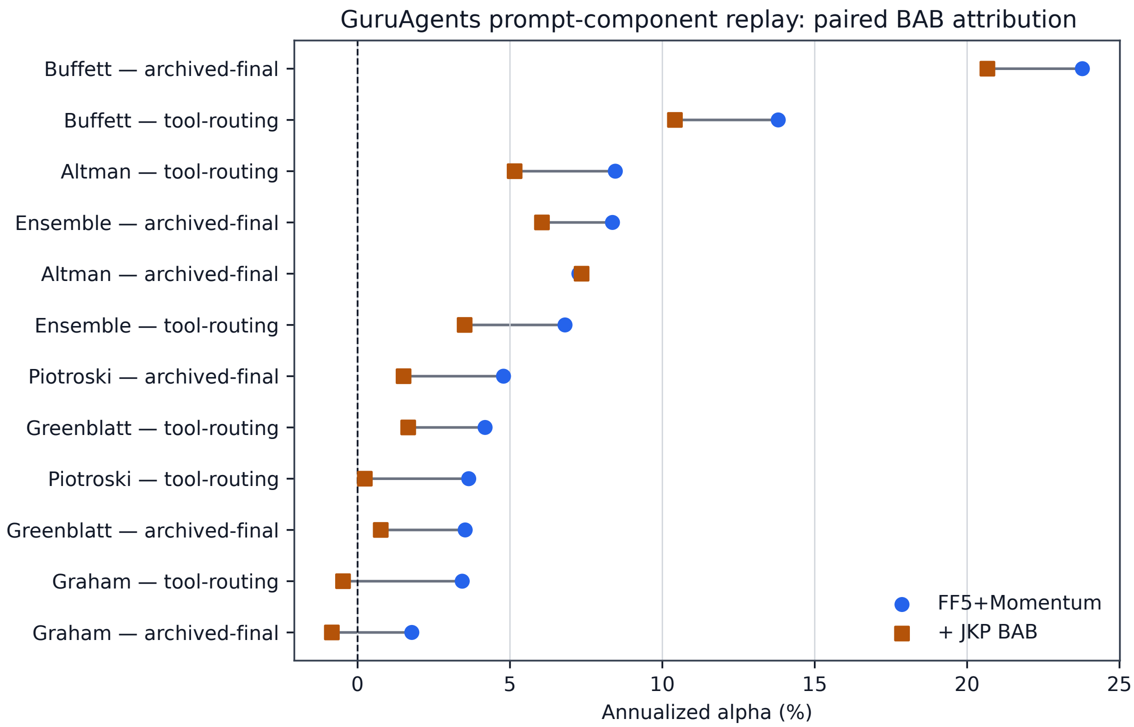

Figure 1: Paired attribution for the 12 long-archive GuruAgents prompt-decision replay paths over the common 33-month window, net of the paper-declared 1 bp one-way costs. Each line joins a path's annualized intercept under official FF5 plus momentum (circle) to the intercept after adding JKP BAB (square). The pairing holds returns and months fixed; it does not make the source data or full agent native.

grades are A0/B0/C15/D33/U2. Its regressions describe researcher-created mappings and cannot adjudicate native-agent performance. The primary formula census instead passes 3/3 strict grade B because QuantEvolver's pinned release supplies both the candidate set and executable evaluator. The nine additional EFS, Alpha-Jungle, and QuantAgent rows remain mixed-fidelity diagnostics: five are B-conditional and four C-conditional, so none enters the primary denominator.

The GuruAgents replay preserves a larger share of the mechanism because the prompts, stated model family, temperature, and deterministic tool transcripts are public. It still conditions on archived observations and a compiled Nasdaq universe, and it does not recreate the source data vendor, original provider or model snapshot, original API state, or authors' exact portfolio path. Its 33-month factor overlap is enough for small OLS models but too short for unrestricted JKP132 OLS. One familywise-positive Buffett result and broad median BAB attenuation should therefore be reported together.

None of the analyses reveals whether an LLM memorized, retrieved, independently rediscovered, or arrived by another route at a familiar factor. Nor do they rank the scientific quality or practical usefulness of the 98 screened works. Formula discovery, search efficiency, interpretability, and decision support can be valuable even when a particular component has familiar factor exposure.

Multiplicity correction addresses sampling variation only within the stated realized families. It does not correct publication selection or the choice among benchmark specifications [2]. The primary candidate set is an exhaustive pinned-release census and its implementation was fixed without reference to returns. An outcome-blind synthetic conformance harness executes the exact pinned evaluator and matches 1,296 scores plus all 105 eligible portfolio timestamps; no market outcomes enter that check. Sasha completed the D07 three-row owner attestation on August 11, 2026, confirming all five fidelity checks and grade B for each row. This human review remains separate from automated conformance. The disclosed components should therefore be independently rerun with native-frequency data before stronger performance claims are made.

5 Reproducibility

The evidence bundle separates three objects. The strict audit records each legacy mapping's source anchor, implemented signal, grade, and reason. The primary formula bundle pins source-file hashes and records its exhaustive candidate census, expressions, evaluator rules, fidelity ledger, formation holdings, return paths, benchmark results, Holm values, and closest JKP factors; a validator reconstructs the full evidence and also supports a compact public package. The older 12-formula mixed-fidelity bundle remains explicitly nonprimary. The GuruAgents bundle records all 190 model outcomes, holdings, monthly returns, traded notional, official factor panels, fitted values, residuals, penalties, loadings, hashes, software versions, and licensing cautions.

No orders were transmitted. Results are retrospective research evidence, not investment advice. Generative AI assisted literature organization, code drafting, diagnostic review, and prose editing; the human author remains responsible for source selection, operator conventions, statistical specifications, citation verification, and all reported claims.

6 Conclusion

The original 50-strategy common-task layer does not support a claim that JKP132 absorbs financial LLM agents. A strict review finds A0 faithful systems, B0 faithful disclosed components, C15 materially changed but recognizable ideas, D33 inconsistent mappings, and U2 unavailable sources. We retain that layer as a construction diagnostic and not as the paper's main empirical test.

The repaired evidence is narrower and more informative. An exhaustive pinned-release census passes 3/3 strict grade B (100%) for disclosed QuantEvolver evaluator components. Median residuals are non-monotone across benchmarks (+1.2250%, +0.2549%, −0.4009%, and +0.6713%); one component is Holm-positive under FF3 and none under the broad JKP model. The GuruAgents prompt-decision replay shows that prompting changes holdings and that adding BAB reduces matched median FF5-plus-momentum alpha from 6.12% to 2.87%, while one short-window Buffett path remains Holm-positive in the identified OLS models. This does not prove either universal novelty or universal absorption.

The methodological recommendation is concrete: alpha claims for financial LLM systems should identify the empirical object, release dated outputs, preserve disclosed formulas or prompts, report operator semantics, separate component tests from system replications, include costs and broad factor attribution, and state the multiplicity family exactly.

7 Construction Diagnostic: Legacy JKP-Built Proxies

For transparency, this section retains the earlier 50-strategy ladder after changing its interpretation. The family contains researcher-authored monthly U.S. mappings for 40 retained papers. Thirty-seven were previously described as favorable in-spirit translations and 13 as source-grounded, but strict formula-level review grades the latter C8/D5. Candidate returns, 10 bp one-way costs, and 126 evaluation months are fixed across benchmark specifications. CAPM, FF3, and FF5 plus momentum use rolling OLS; the market-plus-JKP132 specification leaves six base factors unpenalized and ridge-controls the remaining JKP factors. Newey–West statistics use lag four. Holm is applied across all 50 mappings separately within each benchmark, conditional on the post-hoc mappings; it is not across benchmark specifications and does not cover mapping discretion.

Median annualized residual return in this construction family moves from 9.97% under CAPM to 3.17% under FF3, 2.69% under FF5 plus momentum, and -1.69% under the broad model. The corresponding positive Holm counts are 6, 1, 2, and 0 out of 50. The median nearest-factor absolute correlation is 0.81, and 47 mappings exceed 0.50. These numbers show that the implemented JKP-built composites are strongly spanned by the library used to construct them. They do not transfer to the cited agents.

The secondary 14-target code inventory also reproduces zero native agents. Its blockers and one QuantEvolver seed adaptation remain in the artifact ledger, but do not enter a native-performance denominator. International proxy results are likewise excluded from conclusions because 35 paths cross nonpositive NAV and are dominated by unverified extreme returns held short.

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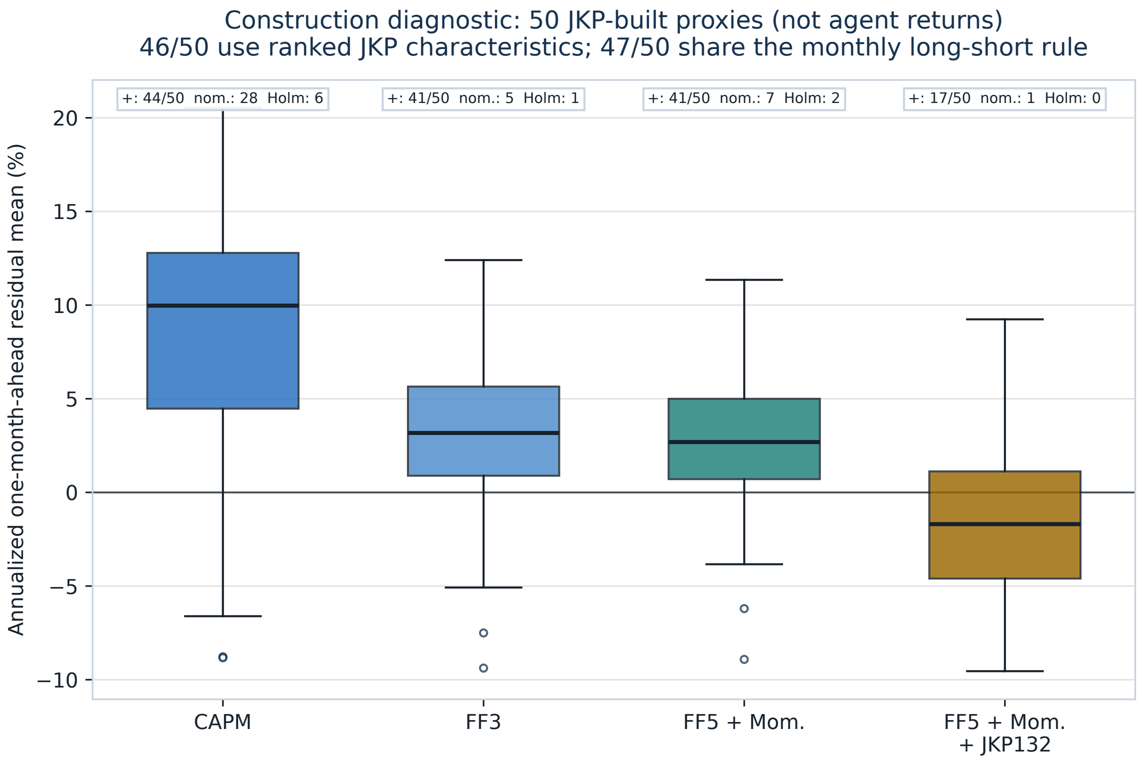

Figure 2: Construction diagnostic for 50 low-fidelity researcher-authored mappings, not a replication result for the cited papers. Returns, costs, and months are matched across models. Annotations report positive estimates, positive two-sided HAC tests at 5%, and positive Holm-adjusted tests within the 50-mapping family for each benchmark. The broad attenuation is expected to reflect, in part, that 46 scores are built from ranked JKP characteristics.